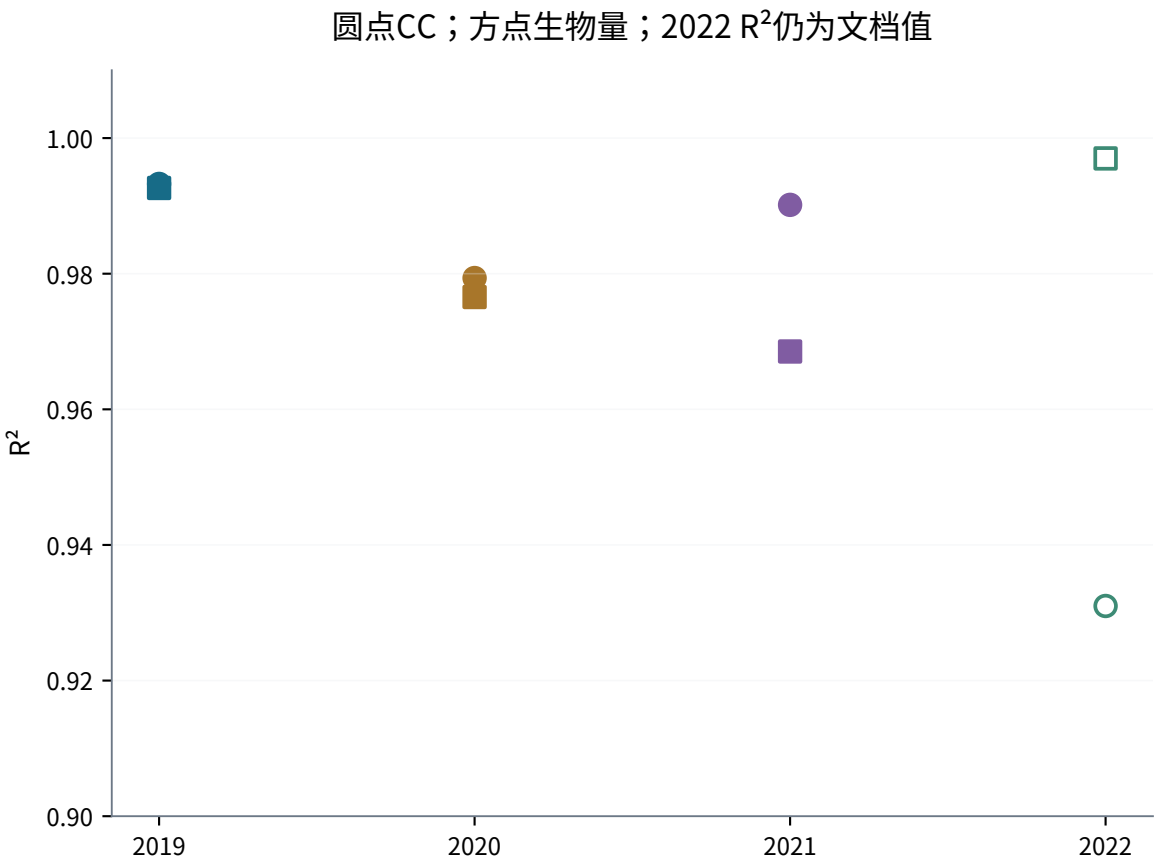
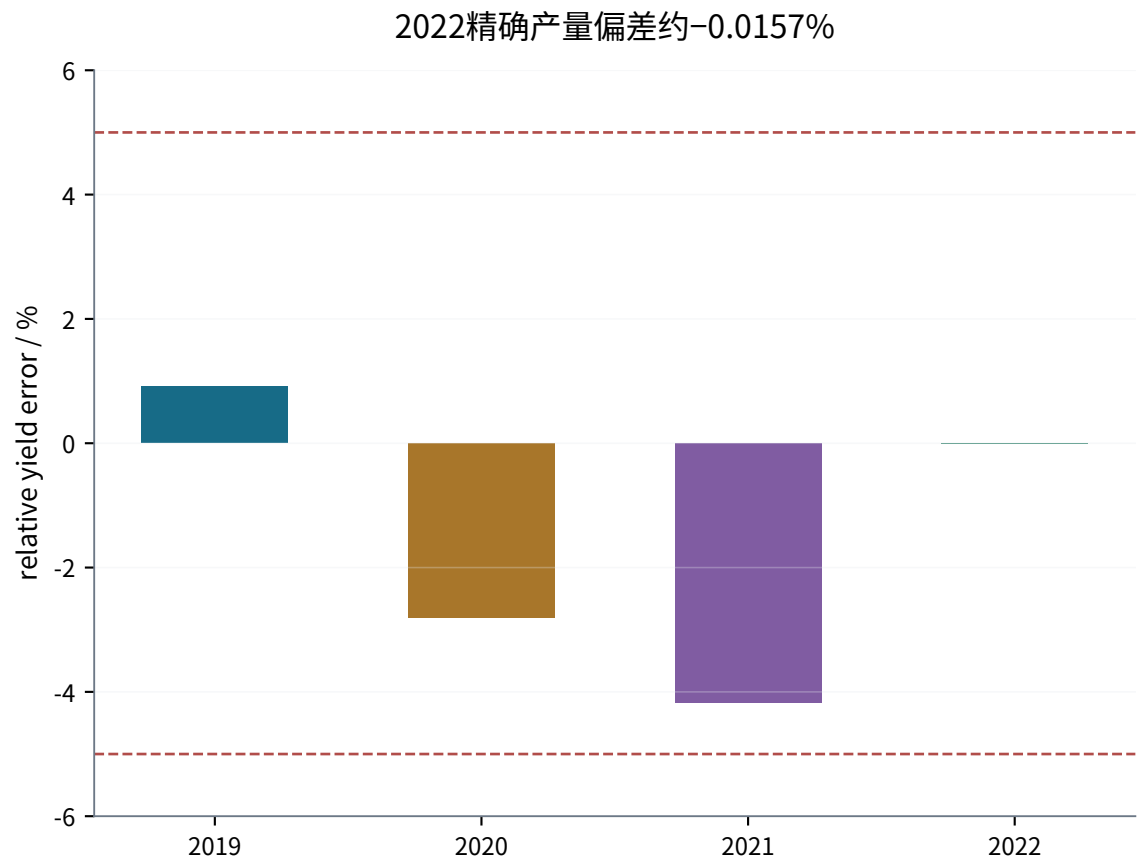
原链已补齐，配对序列证据仍需分别标注





2022空心R²仍为原Word中的汇总值，未从配对表重算；产量偏差与RMSE已更新为新JSON精确值。±5%为项目容许幅度。